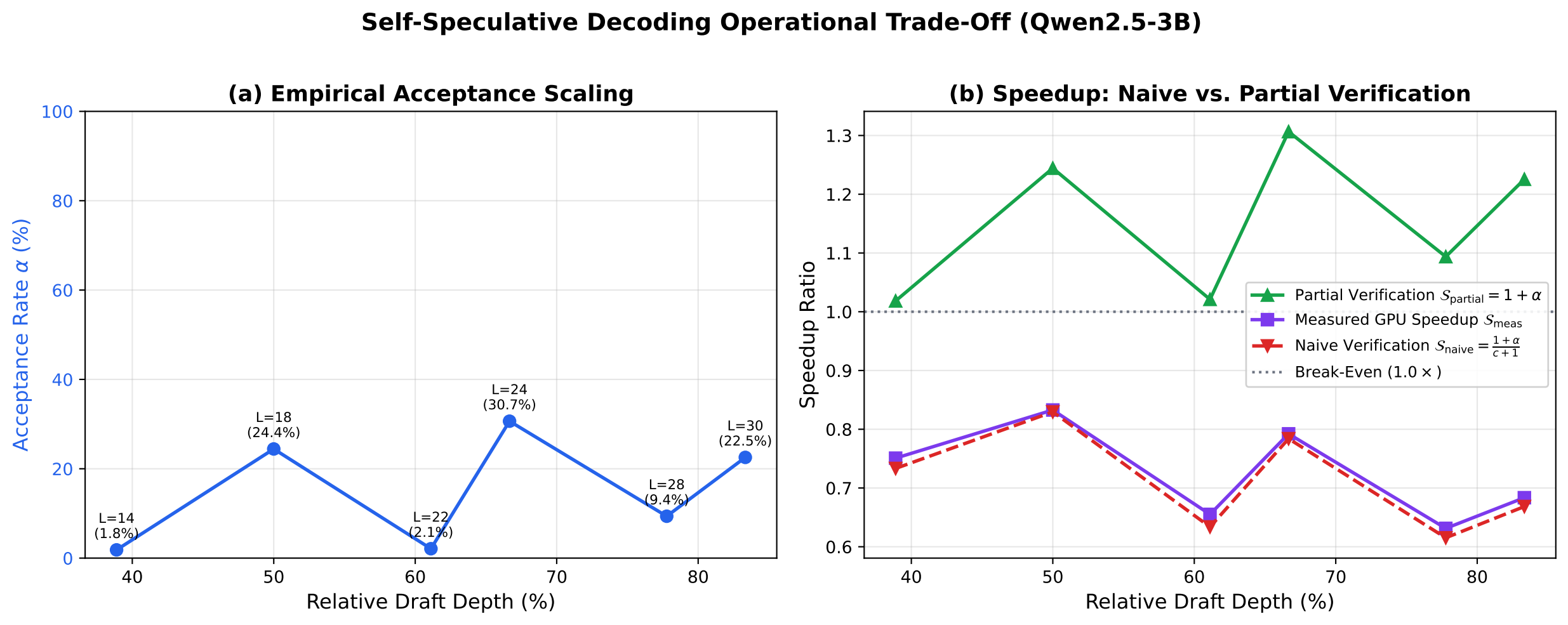
Self-Speculative Decoding Operational Trade-Off (Qwen2.5-3B)
(a) Empirical Acceptance Scaling
Acceptance Rate α (%)
Relative Draft Depth (%)
L=14
(1.8%)
L=18
(24.4%)
L=22
(2.1%)
L=24
(30.7%)
L=28
(9.4%)
L=30
(22.5%)
(b) Speedup: Naive vs. Partial Verification
Speedup Ratio
Relative Draft Depth (%)
Partial Verification $\mathcal{S}_{\text{partial}} = 1 + \alpha$
Measured GPU Speedup $\mathcal{S}_{\text{meas}}$
Naive Verification $\mathcal{S}_{\text{naive}} = \frac{1+\alpha}{c+1}$
Break-Even (1.0 ×)